

need to find out an alternative to hydrogen electrolyser.

One simple alternative is to have second set of batteries. This can be a chemical battery or a gravity battery. Gravity batteries will have very long life as compared to chemical batteries.

A cow-shed can be established near the farm. As mentioned earlier, the land below the PV panels can be used for farming. In the proposed scheme, the land is used to produce fodder (fresh grass) for the cattle. After meeting fodder requirement, if still there is spare land available, vegetables and fruits can be grown for additional income.

Regular income can be earned by selling cow's milk. There is also a big demand for cow urine for medical purposes. The cow dung can be used for the biogas plant. The raw biogas generated contains carbon-dioxide and water vapour. Once these two are removed, we get 90% methane, which can be compressed to get bio-CNG.

This compressed bio-CNG can be used for electric power generation during night. The output of this generator is AC, which can be directly fed to the L2 chargers. A rectifier is used to convert AC to DC for the DC superfast charger.

In summer, the PV power generated is more than in winter. Hence, in summer we may have excess bio-CNG in the tanks. Excess bio-CNG can be filled in the vehicles through the CNG dispenser, as shown in Fig. 3.

The PV panels can be installed on roof-top of the cow shed. This will further increase the PV power generation, providing additional income.

Automation

The proposed system is fairly complex. Therefore, complete automation has to be done for proper operation of the system and to maximise return on investment.

Further, all safety systems should be installed for avoiding any untoward incidents.

The solar radiation and weather changes are highly fluctuating. Hence, weather forecasts for next few days should be provided as input to the automation system. Similarly, the pattern of demand for EV charging can be studied and fed as input to the system.

During prolonged spell of rains, the PV power may not be sufficient for meeting the power demands. In that case, for few days, the normal CNG could be used as a backup fuel. Use of artificial intelligence and machine learning program will be very useful for optimal running of the system.

Implementation

The proposed scheme can be implemented in two steps. The first step is to check technical feasibility. A pilot plant can be established and run for 2-3 years to gain insight into the operation of the system. It will generate very useful data.

In the second step, full-scale plant can be built, based on the learnings from the pilot plant. Optimisation can be done based on the data available, in order to reduce the cost of the system.

The cost of implementation will be on the higher side due to the complex nature, but this integrated system will offer multiple benefits. The return on investment will be much higher, which will reduce the payback period.

Some advantages of the system are:

- Off-grid operation reduces burden on grid on daily basis
- Off-grid operation is extremely useful during emergency grid outages
- Shorter cables and less transmission losses
- Carbon-neutral system
- Self-sufficient with minimal movement of material

Some sources of revenue generation are:

- EV charging service
 - Bio-CNG dispensing
 - Sale of milk
 - Sale of organic manure
 - Sale of cow urine for medical usage (can even be exported)
 - Can claim carbon credits
- And some avenues of the savings from the project are:
- Vehicles used for selling milk and manure get free charging or free CNG
 - Free electricity for the complex
 - Conservation of water because of agri-voltaics
 - Extra power generation because of lower operating temperature of panels
 - Water used for cleaning the panels goes for irrigation
 - Fodder for cattle grown in-house; less cost and no transportation
 - Vehicle owners waiting for EV to charge can visit cow shed and feed the cows

Conclusion

Along the highways there is enough land available for implementing the proposed scheme. An integrated system offers many more advantages. The return on investment is high. Most of the inputs for running the system are available inhouse. Hence there is minimal movement of material.

Except green hydrogen, all other technologies can be easily implemented. The equipment required for the system is available within the country. Thus, a pilot plant can be started with most of the equipment procured locally.

In future, more advanced technologies, such as fuel cells, could be used for more efficient and silent operation. This off-grid system will be very useful when there is a sudden grid failure. It will continue to provide charging for two- and four-wheelers. **EFY**